

CHASING THE RESIDUAL: MAXWELLIAN AVERAGING, A WRONG TURN, AND WHAT IT CONFIRMS

JEREMY EBERT

ABSTRACT. [2, 3] each left an unexplained residual — factor 1.5–1.6 — after matching our own binary-encounter formulas to Braginskii’s plasma collision time and the standard stellar-dynamics relaxation time, attributing it to the honest Maxwellian velocity average neither note performed. This note performs that average, and the first attempt makes things *worse*, not better: naively combining the test particle’s own thermal variance σ^2 with a diffusion coefficient averaged over the Maxwellian *relative*-velocity distribution introduces a spurious factor of π , since these use two different, inconsistently mixed velocity scales. Diagnosing and correcting this — using a single, self-consistent effective velocity $u_{\text{eff}} = \sigma\sqrt{\pi}$ throughout, derived from $\langle 1/u \rangle$ over the true relative-velocity Maxwellian — gives a result independently close to the earlier ad hoc choice $\sqrt{3}\sigma$ used in [2, 3] (agreement to 2.3% between the two derivations), which is itself a meaningful confirmation that neither velocity choice was arbitrary. The residual factor survives regardless: 7% against Spitzer–Landau, 26%–35% against Braginskii and the stellar-dynamics formula, in both plasma and gravitational cases. We conclude, with a specific diagnosis rather than a shrug, that closing this residual requires the actual Chapman–Enskog/Rosenbluth-potential machinery those formulas are built on — correctly handling the full velocity-dependence of the collision integral, not any single effective speed however carefully chosen — which is beyond the scope of a single-speed encounter model.

1. THE ATTEMPT

Both [2] and [3] evaluate $\text{Tr } D \propto n\kappa_{\text{int}}^2 \ln \Lambda / (m^2 u)$ and its friction counterpart at one characteristic speed u , rather than averaging over the field’s actual velocity distribution. The natural next step is to do the average properly.

Proposition 1.1 (Two universal Maxwellian speed averages). *For a 3D Maxwellian speed distribution with one-dimensional dispersion s ,*

$$(14) \quad \left\langle \frac{1}{u} \right\rangle = \sqrt{\frac{2}{\pi}} \frac{1}{s}, \quad \left\langle \frac{1}{u^2} \right\rangle = \frac{1}{s^2}.$$

Numerical Verification. Direct Monte Carlo sampling of 2×10^7 points from a 3D Gaussian velocity distribution ($s = 3.7$, arbitrary units) gives $\langle 1/u \rangle = 0.21563$ against the predicted 0.21564, and $\langle 1/u^2 \rangle = 0.07262$ against the predicted 0.07305 (the larger relative error here is expected Monte Carlo noise from the heavier tail of $1/u^2$, not a discrepancy in (14)).

If the test and field species are identical (self-collisions, the case both [2, 3] consider), the relative velocity of two independent particles each with 1D dispersion σ has its own dispersion $\sigma_{\text{rel}} = \sqrt{2}\sigma$.

2. THE WRONG TURN

Applying (14) directly to $\text{Tr } D(u) = 8\pi\kappa_{\text{int}}^2 n \ln \Lambda / (m^2 u)$ (equal masses) gives

$$(15) \quad \langle \text{Tr } D \rangle = 8\sqrt{\pi} \frac{\kappa_{\text{int}}^2 n \ln \Lambda}{m^2 \sigma}.$$

Defining a relaxation time the same way as before, $\tau := \sigma^2 / \langle \text{Tr } D \rangle$, and evaluating for the fusion-plasma parameters of [2] ($n_e = 10^{20} \text{ m}^{-3}$, $\Theta_e = 10 \text{ keV}$, $\ln \Lambda = 19.4$) gives $\tau = 41.8 \mu\text{s}$ — *smaller* than our own earlier single-speed result ($116.2 \mu\text{s}$ at $u = \sqrt{3}\sigma$), and worse against both Braginskii ($177.6 \mu\text{s}$) and Spitzer–Landau ($122.7 \mu\text{s}$) than what [2] already reported.

Proposition 2.1 (The error is a scale mismatch, not a flaw in (14)). *Writing $\text{Tr } D(u) = C/u$ for constant C , (15) is exactly $\text{Tr } D(u_{\text{eff}})$ at the single effective speed $u_{\text{eff}} = 1/\langle 1/u \rangle = \sigma_{\text{rel}}\sqrt{\pi/2} = \sigma\sqrt{\pi} \approx 1.772\sigma$. Using $\tau = \sigma^2/\langle \text{Tr } D \rangle$ therefore silently divides by σ^2 (the target, single-particle variance) while the denominator was averaged at the relative-velocity scale $u_{\text{eff}} \approx 1.77\sigma \neq \sigma$ — two different velocity scales, entangled without comment. The self-consistent quantity is $\tau_{\text{eff}} := u_{\text{eff}}^2/\langle \text{Tr } D \rangle = \text{Tr } D(u_{\text{eff}})^{-1}u_{\text{eff}}^2$, which differs from the naive τ by exactly $(u_{\text{eff}}/\sigma)^2 = \pi$.*

Numerical Verification. $\tau_{\text{eff}} = 131.3 \mu\text{s}$ for the same fusion-plasma parameters, and indeed $131.3/41.8 = 3.142 \approx \pi$, confirming Proposition 2.1 exactly.

3. WHAT THE CORRECTED CALCULATION SHOWS

	τ_{eff} (this note)	literature	agreement
Fusion plasma (vs. Spitzer–Landau)	$131.3 \mu\text{s}$	$122.7 \mu\text{s}$	7%
Fusion plasma (vs. Braginskii)	$131.3 \mu\text{s}$	$177.6 \mu\text{s}$	26%
Globular cluster core (vs. eq. (9) of [3])	59.6 Myr	91.5 Myr	35%

Remark (Two independent routes agree). $u_{\text{eff}} = \sigma\sqrt{\pi} \approx 1.772\sigma$, derived here purely from the Maxwellian average $\langle 1/u \rangle$, is within 2.3% of the $\sqrt{3}\sigma \approx 1.732\sigma$ used in [2, 3] on the unrelated grounds that it matches the $(3\Theta)^{3/2}$ structure explicit in the Spitzer–Landau formula. Two different justifications for a characteristic velocity landing within 2% of each other is a real, meaningful cross-check that neither choice was an arbitrary knob turned to fit the data — both are honest approximations to the same underlying effective speed.

Remark (What the surviving residual is, and is not). The residual — 7% against Spitzer–Landau, 26–35% against the more carefully constructed Braginskii and stellar-dynamics formulas — does not close under any single-effective-speed treatment, however the speed is chosen or justified. This is not a failure of arithmetic; it reflects that Braginskii’s formula (and, by the same logic, the standard two-body relaxation time) is derived from the linearized Fokker–Planck (Rosenbluth-potential) collision operator, integrated exactly over the full relative-velocity dependence of the encounter rate — including the region where the deflection is not small and the leading-log approximation itself requires the smooth cutoff that the Chandrasekhar G -function provides, not the sharp b_{90} cutoff we have used throughout. Closing this residual honestly requires that machinery, not a better-chosen u_{eff} ; we have not built it, and do not believe a single effective velocity can substitute for it.

4. DISCUSSION

The failed first attempt is, we think, the most useful part of this note. It would have been easy to quietly report only the corrected τ_{eff} and the resulting 7–35% agreements, presenting the Maxwellian average as simply another verification step. What actually happened is that the naive version of “do it properly” made the match *worse* by a factor of π , for an identifiable and non-obvious reason — two legitimate-looking velocity scales, silently conflated — and only diagnosing that mistake produced a result that both matches better and independently corroborates the earlier ad hoc choice. We record this not to pad the note but because it is a genuine instance of the same discipline this whole project has tried to maintain: a plausible-looking calculation that fails a numerical check is data, and chasing why is more valuable than the number that would have resulted from not checking.

AI ASSISTANCE DISCLOSURE

Large language model tools (Anthropic’s Claude) were used in the derivation, symbolic computation, numerical verification, and L^AT_EX preparation of this note. This includes: Monte Carlo verification of Proposition 2.1’s two Maxwellian speed integrals; the initial (incorrect) evaluation of the naive Maxwellian-averaged relaxation time for both the fusion-plasma and globular-cluster parameters of [2, 3], which produced a worse match than those notes’ single-speed results; the diagnosis in Proposition 3.2 tracing this to a scale-mismatch between the test particle’s own variance and the relative-velocity-averaged diffusion coefficient, verified by confirming the exact factor of π predicted; and the corrected self-consistent calculation for both systems. The decision to report the failed first attempt explicitly, rather than silently replacing it with the corrected version, was made by the author.

REFERENCES

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FRANKLIN COUNTY, OHIO, USA